

Municipal Law Skill

for Autonomous Agents

Deterministic, citation-backed municipal law retrieval — no LLM in the loop

NANDA Town · MIT Reality Hackathon 2026

<https://nanda-municipal-laws.vercel.app>

AI agents hallucinate municipal law

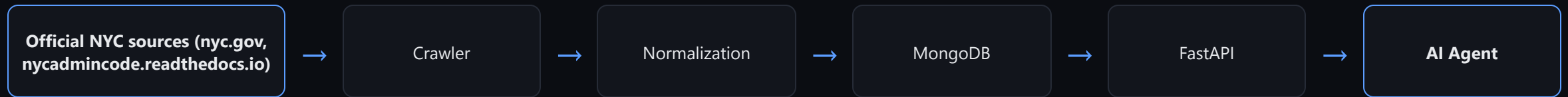
- ▶ Popular internet myth: "NYC caps backyard chickens at 6 hens"
- ▶ The real statute (§161.19) states no such number — it only prohibits roosters, ducks, geese, and turkeys, with narrow exceptions
- ▶ Agents relying on an LLM's training data repeat exactly this kind of confident, wrong answer
- ▶ Legal citations need to come from the actual code — not a model's memory

The capability

“A deterministic municipal law skill that any AI agent can invoke to obtain grounded, citation-backed legal evidence from the complete NYC Administrative and Health Codes.”

- ▶ No LLM calls, ever — nothing for the service itself to hallucinate
- ▶ Same query → same citations → same ordering, every time
- ▶ The calling agent performs the reasoning; this service only supplies evidence

Architecture



The service never generates legal advice — only official text and citations.

The complete corpus, not a sample

NYC Administrative Code

32

Titles

4,781

Sections

NYC Health Code

36

Articles

501

Sections

668 source documents · 10,702 searchable chunks total

Agent-facing APIs — Legal Determination & Search

Method	Endpoint	Use case
POST	/is_action_allowed	Headline capability — yes/no legality check on a described action. Returns allowed, conditions, citations, reasoning, confidence.
POST	/search	General keyword lookup across the entire corpus — filterable by title, chapter, document_type, agency, or topic.
POST	/sections/{id}/term_map	Highlight exactly where search terms occur within a section — built for search-results/demo UI rendering.

Agent-facing APIs — Lookup & Cross References

Method	Endpoint	Use case
GET	/sections/{id}	Exact section retrieval by number — full untruncated text plus a deterministic structural summary.
GET	/sections/{id}/related	Resolve a section's own cross-references into their citations — a one-hop citation graph.
GET	/documents/{id}	Metadata for a source document (an Admin Code chapter or Health Code article).
GET	/documents/{id}/chunks	Every section belonging to one document, in order.

Agent-facing APIs — Penalties & Permits

Method	Endpoint	Use case
POST	/penalties	Find sections flagged as mentioning a penalty or fine — for enforcement-related questions.
POST	/permits	Find sections flagged as requiring a permit or license.

Administration & operational endpoints

Kept separate from the agent-facing retrieval surface above — not part of the reasoning loop

Method	Endpoint	Use case
GET	/health	Liveness check — reports MongoDB reachability.
GET	/version	Deployed app version, plus corpus freshness: last-ingested date and age in days.
GET	/pubkey	Ed25519 public key for verifying a signed response's provenance offline.
POST	/ingest	Fetch, parse, and persist new source documents. Operational only, API-key gated — not for agent use.

Story 1 — A resident

"Can I keep backyard chickens?"

1. Agent calls POST /is_action_allowed

2. allowed: true (confidence: medium) — citing §161.19

3. Agent answers: "Yes, but roosters are prohibited citywide"

Story 2 — A food vendor

“What permit do I need to operate a mobile food cart?”

1. Agent calls POST /permits {"query": "mobile food vending unit license permit"}

2. Top result: \$89.11 “Applications for permits and licenses” (Mobile Food Vending)

3. Agent answers: “You need a vending permit/license under \$89.11 — fees are set by §17-308”

Story 3 — A building owner

“What's the penalty if I dump construction debris illegally?”

1. Agent calls POST /penalties → GET /sections/16-119

2. §16-119 “Dumping prohibited” — misdemeanor, \$1,500–\$10,000 civil penalty, vehicle impoundment

3. Agent answers with real dollar figures, straight from the statute

Engineering: bugs found and fixed, not just claimed

- ▶ Naive substring search: “fine” matched inside “defined” — fixed with word-boundary matching everywhere
- ▶ 36 Health Code PDFs, each formatted slightly differently — parser normalized and verified section-by-section against real text
- ▶ The “6 hens” myth — a regression test locks in the true statute text, forever

Trust & verification

- ▶ Every `/is_action_allowed` and `/search` response is Ed25519-signed
- ▶ `/pubkey` lets any downstream agent verify offline that this service — not a relay or a cache — produced a citation
- ▶ `/version` reports corpus freshness — law goes stale, and callers can know exactly how stale
- ▶ Same query → same citations → same ordering — no randomness, ever

Why NANDA

“This skill lets any autonomous agent in NANDA Town answer municipal-law questions using authoritative legal sources — instead of relying on an LLM's memory.”

- ▶ Composable by design — the last hop in a chain started by another skill (a 311 complaint, a permit application, a code-enforcement lead)
- ▶ Any agent that can call an HTTP endpoint can invoke it, per SKILL.md

Agent-ready. Deterministic. Citation-backed.

4,781 NYC Administrative Code sections

501 NYC Health Code sections

No hallucinations — grounded by citations, every time

[**https://nanda-municipal-laws.vercel.app**](https://nanda-municipal-laws.vercel.app)

<https://github.com/swamx/nanda-municipal-laws>